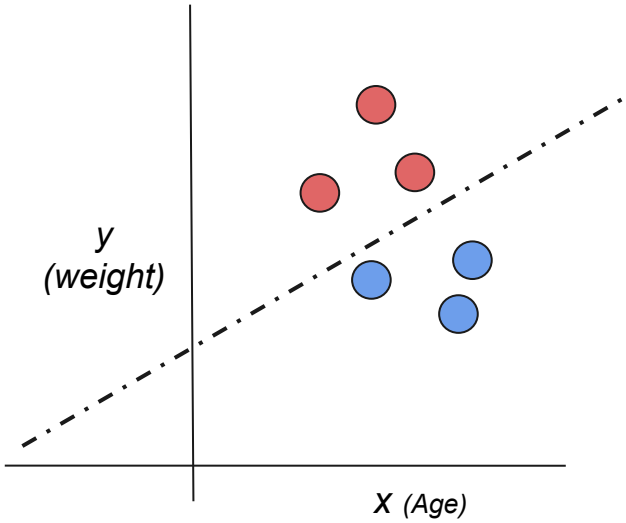
CIS 678 - Machine Learning

Classification Models: Logistic Regression

Classification Models

- *k-NN*
- *Decision Tree*
- **Logistic Regression**
- Random Forest Classifier
- Support Vector Machines (SVMs)
- Boosting Classifiers
- Naive Bayes

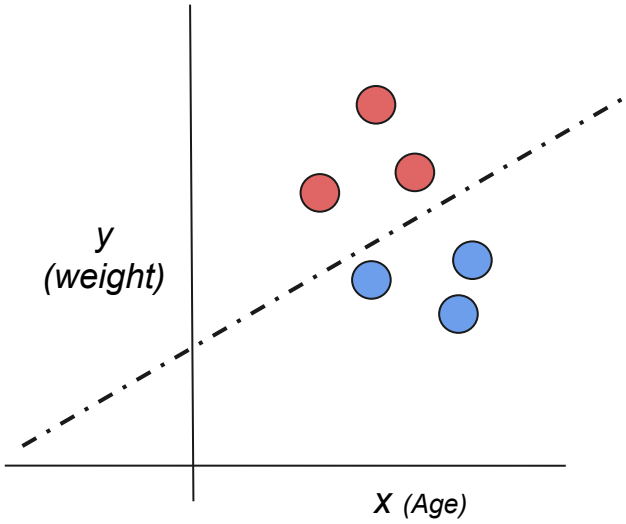
Linear Line as a Threshold



Straight Line as the Separator

$y \in \{underweight(0), overweight\ (1)\}$

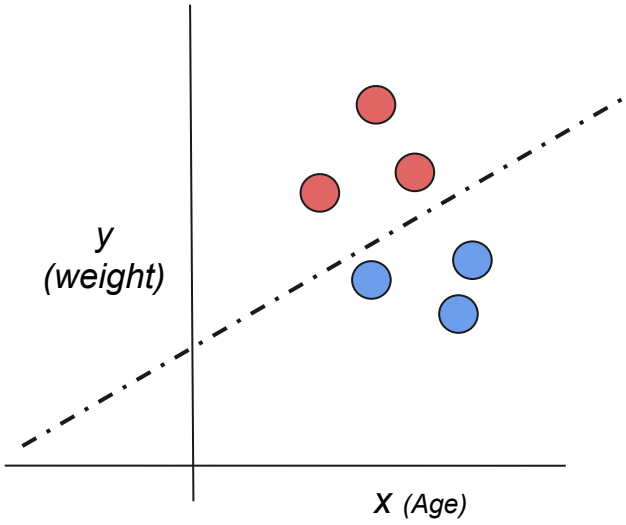
Linear Line as a Threshold



Straight Line as the Separator

$$\boxed{\hat{y} = \beta_0 + \beta_1 x}$$

Linear Line as a Threshold



Straight Line as the Separator

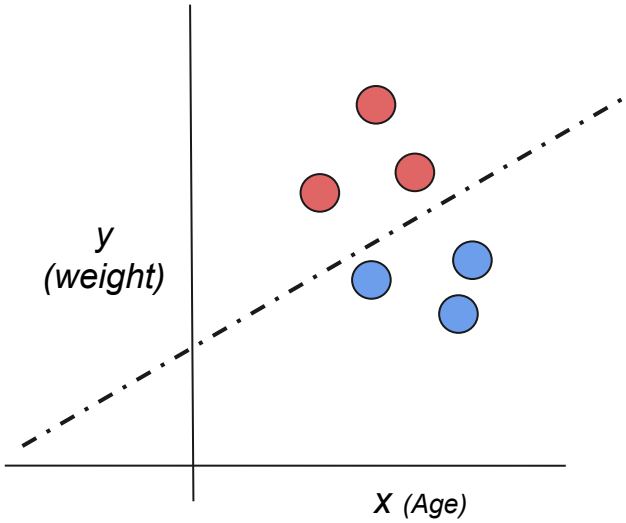
$$\boxed{\hat{y} = \beta_0 + \beta_1 x}$$

Classification Rule

$$\check{y} = \begin{cases} 1, & \text{if} \quad y - (\beta_0 + \beta_1 x) > 0 \\ 0, & \text{otherwise} \end{cases}$$

Logistic Regression

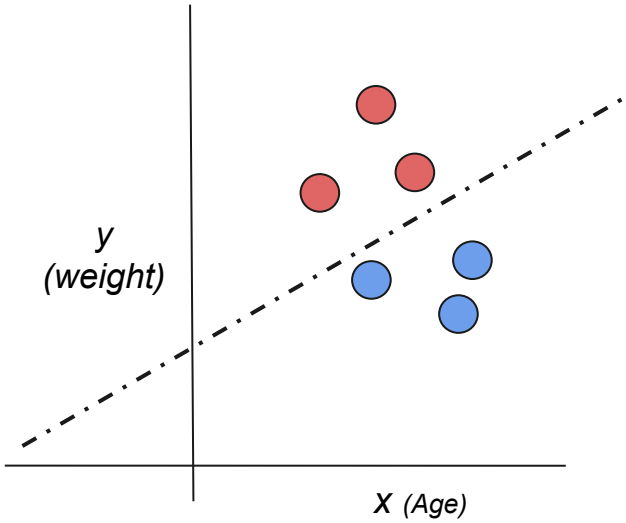
From Regression to Classification



$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Logistic Regression

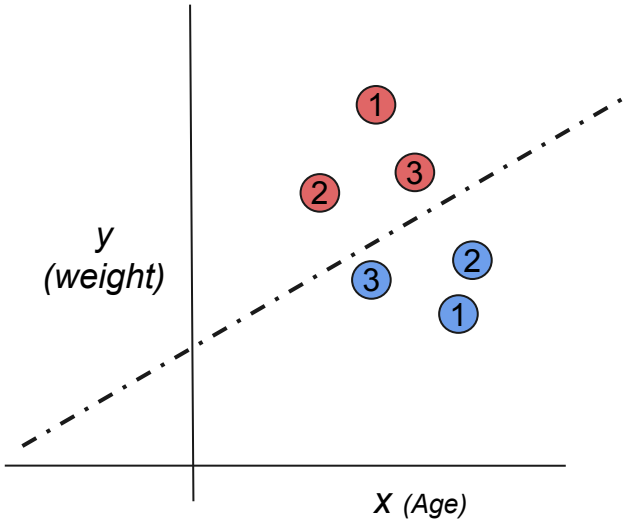
From Regression to Classification



- Relationship to Decision Boundary
  (The dotted line in this case)

Logistic Regression

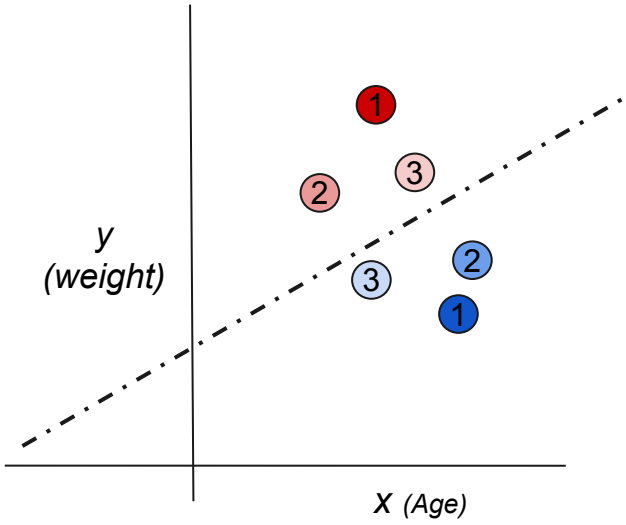
From Regression to Classification



- Relationship to Decision Boundary (The dotted line in this case)
- We index based on their distance

Logistic Regression

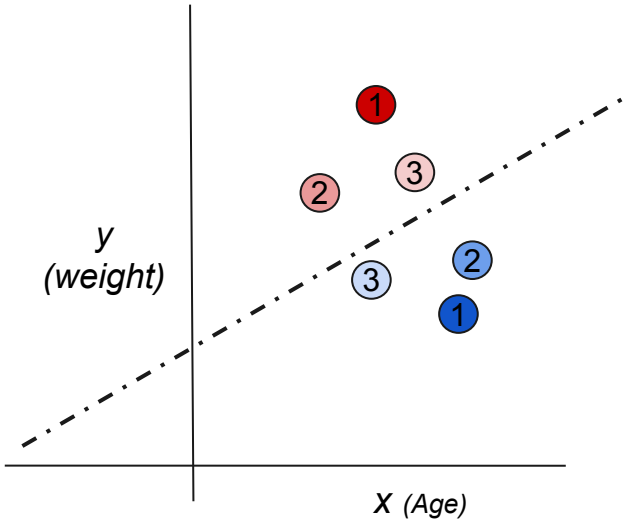
From Regression to Classification



- Relationship to Decision Boundary (The dotted line in this case)
- We index based on their distance
- Also add color code for better understanding

Logistic Regression

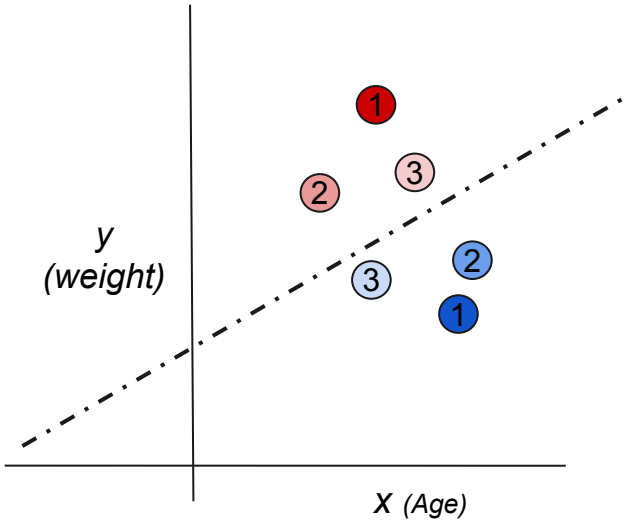
From Regression to Classification



- Relationship to Decision Boundary (The dotted line in this case)
- We index based on their distance
- Also add color code for better understanding
- Let's do feature encoding

Logistic Regression

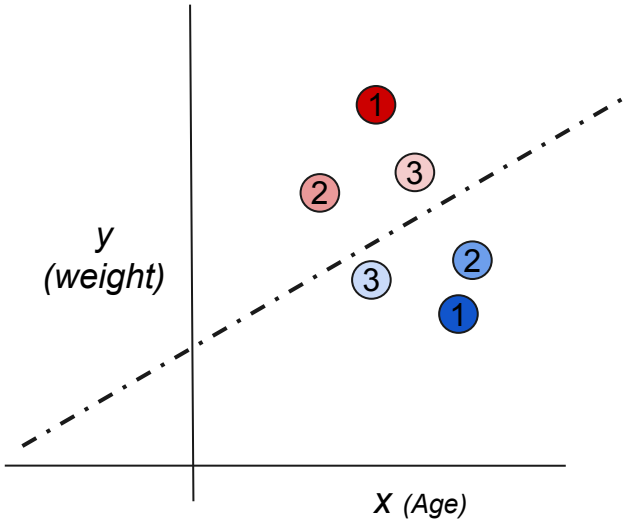
From Regression to Classification



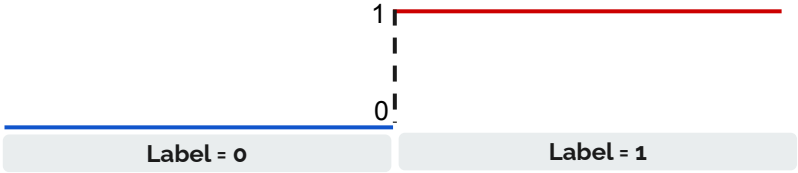
- Let's do feature encoding: ***Binary {0, 1}***

Logistic Regression

From Regression to Classification

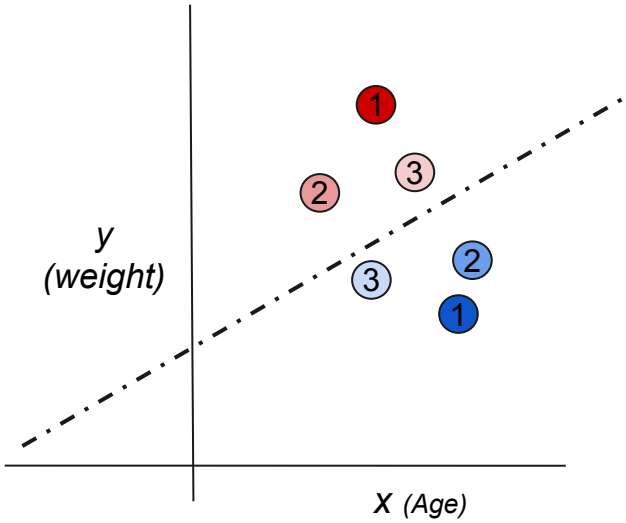


- Let's do feature encoding: ***Binary {0, 1}***

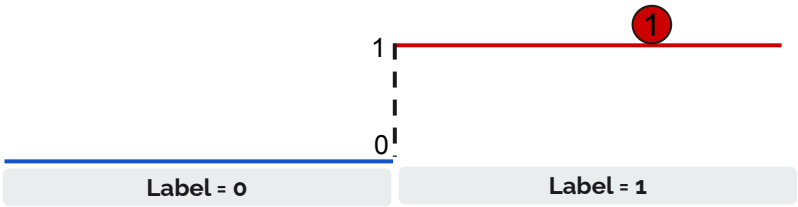


Logistic Regression

From Regression to Classification

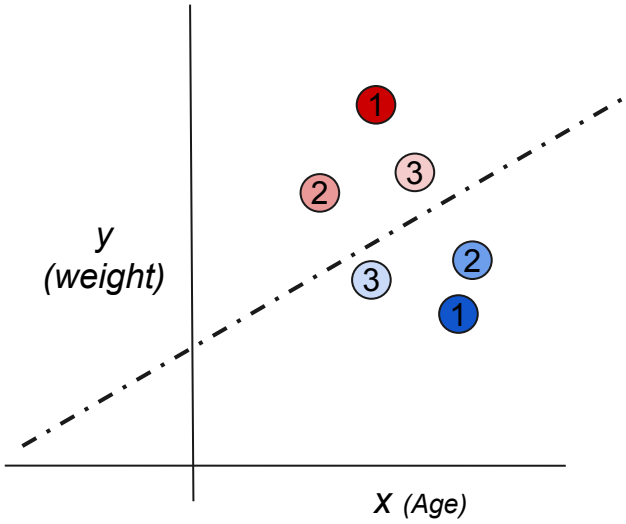


- Let's do feature encoding: ***Binary {0, 1}***

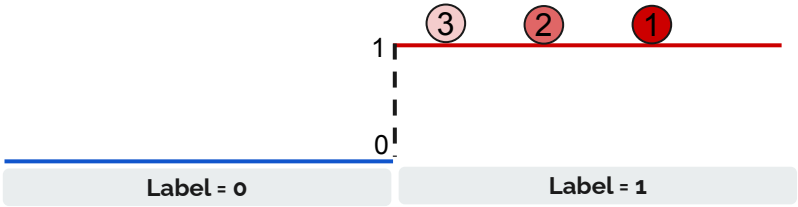


Logistic Regression

From Regression to Classification

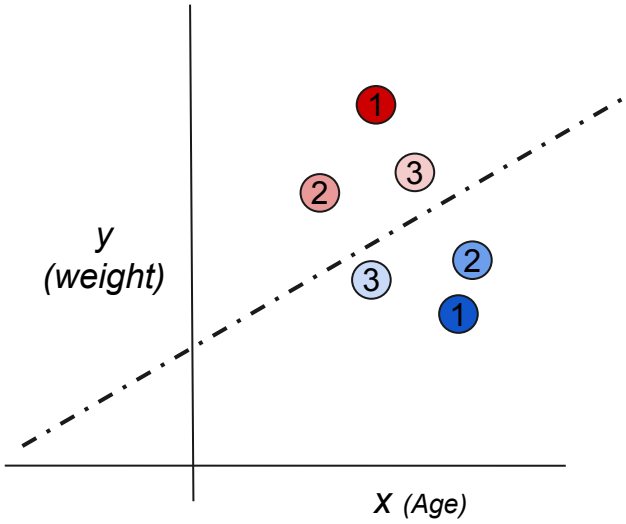


- Let's do feature encoding: ***Binary {0, 1}***

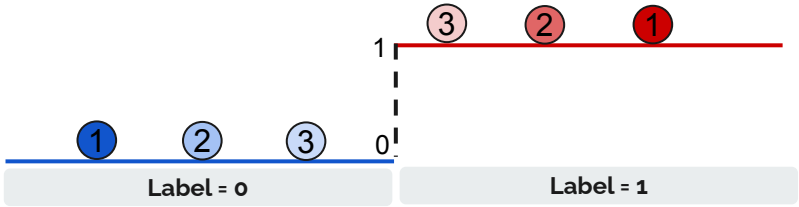


Logistic Regression

From Regression to Classification

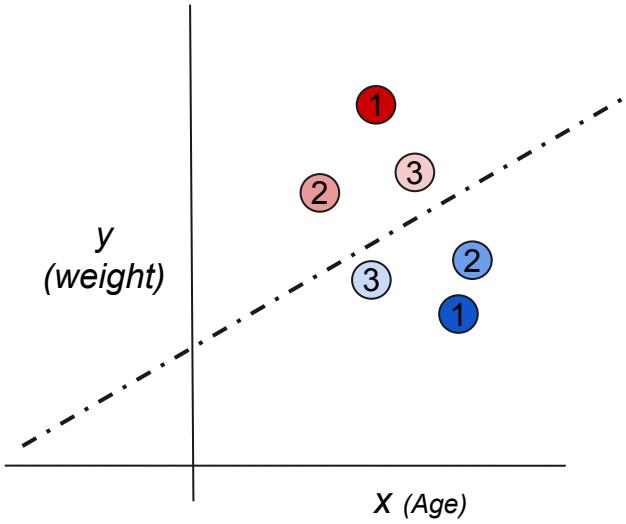


- Let's do feature encoding: ***Binary {0, 1}***

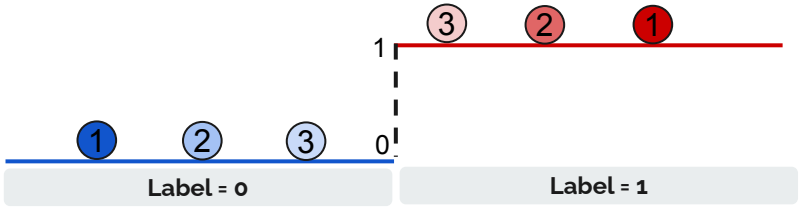


Logistic Regression

From Regression to Classification

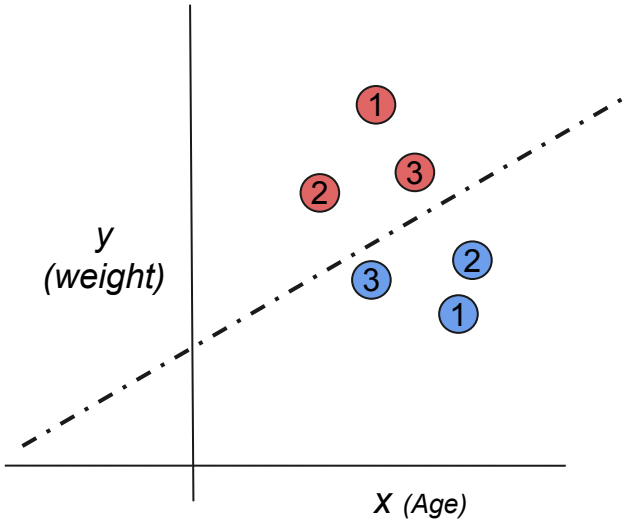


- Let's do feature encoding: ***Binary {0, 1}***
- We have to sacrifice color density (all from the same class have the same label, either 0/1

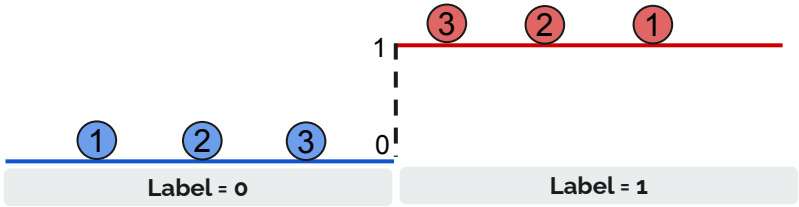


Logistic Regression

From Regression to Classification

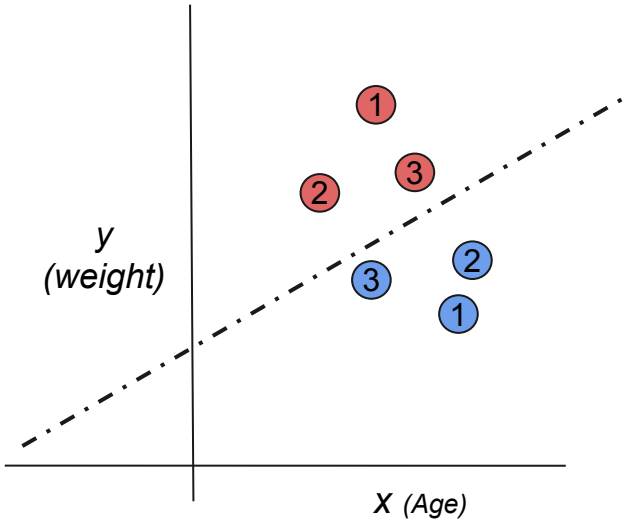


- Let's do feature encoding: ***Binary {0, 1}***
- We have to sacrifice color density (all from the same class have the same label, either 0/1)

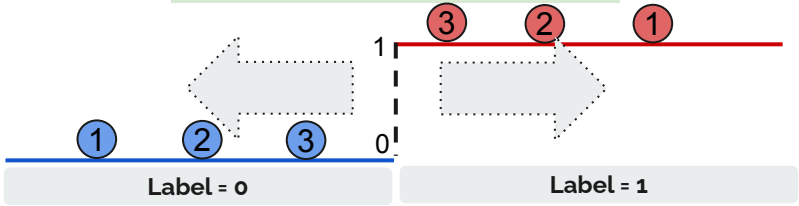


Logistic Regression

From Regression to Classification

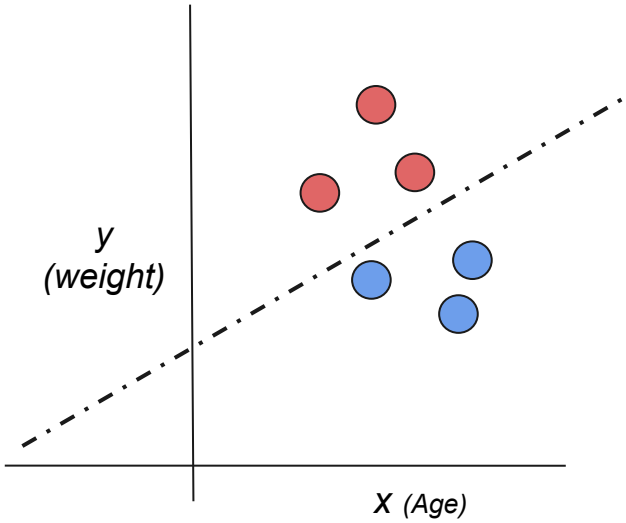


- Let's do feature encoding: ***Binary {0, 1}***
- We have to sacrifice color density (all from the same class have the same label, either 0/1
- We capture the indexing relationship with the distance from the 0-axis.



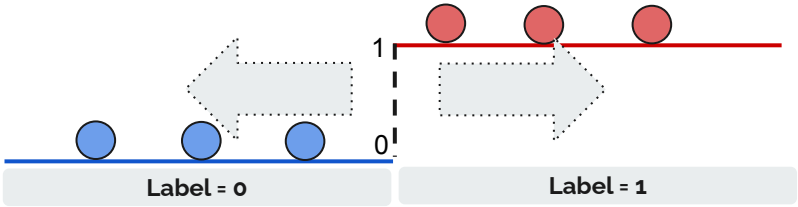
Logistic Regression

From Regression to Classification



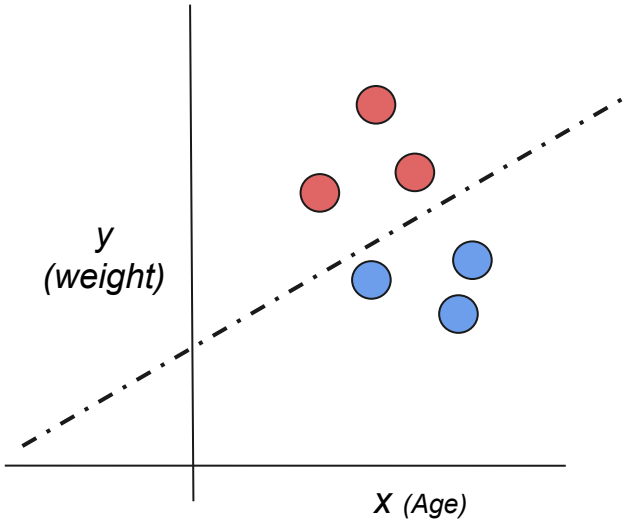
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function;*



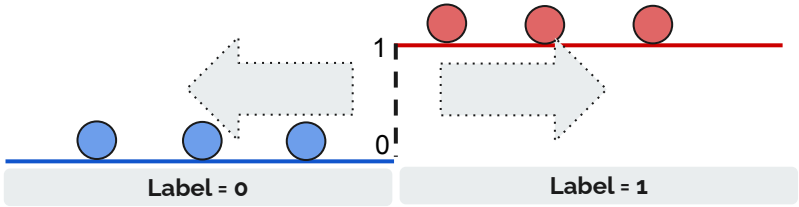
Logistic Regression

From Regression to Classification



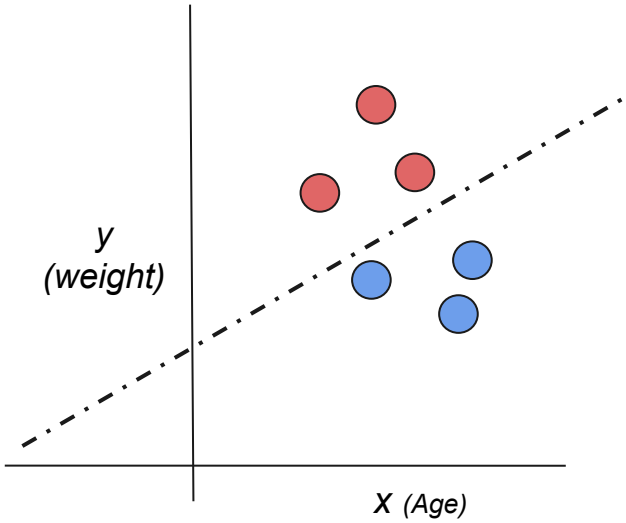
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function; how?*



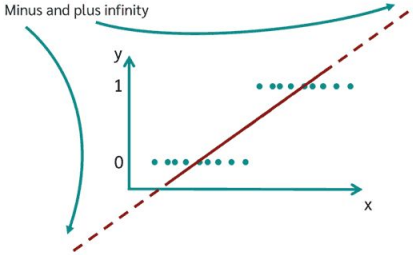
Logistic Regression

From Regression to Classification



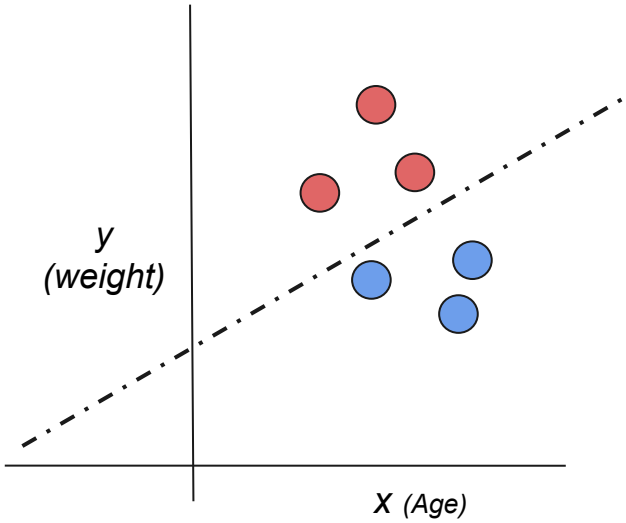
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function; how?*



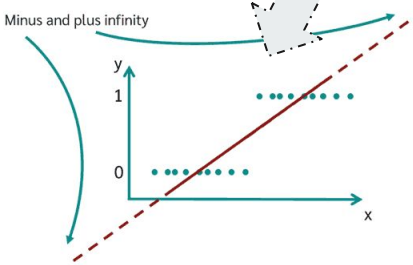
Logistic Regression

From Regression to Classification



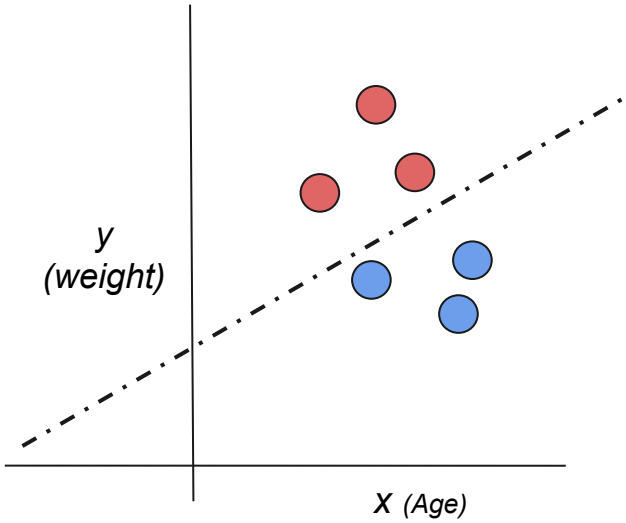
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function. How?*



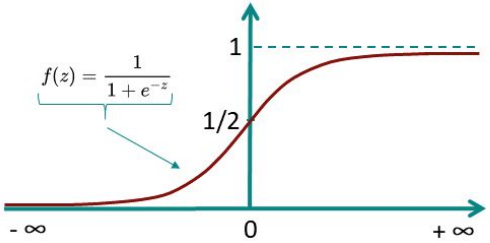
Logistic Regression

From Regression to Classification



$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function; how?*



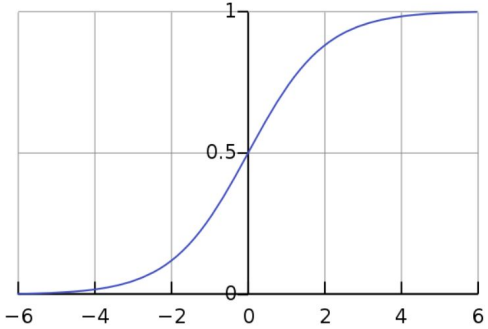
Logistic Regression

- Probabilistic classifier

[Sigmoid function characteristic]

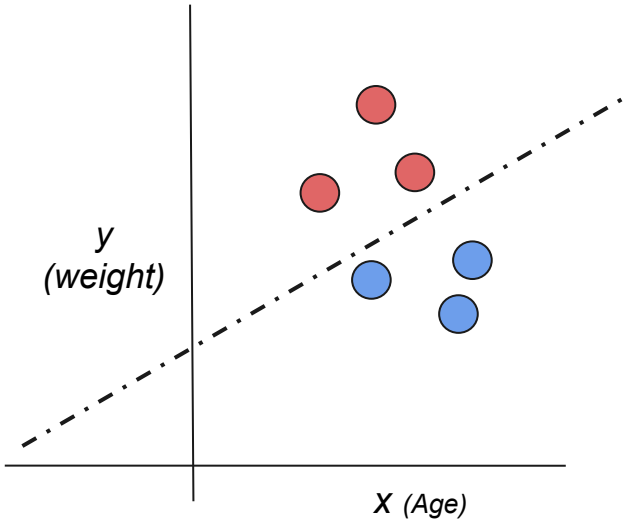
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- Sigmoid function



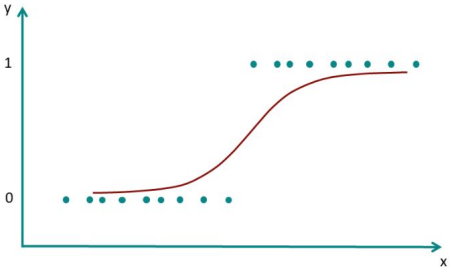
Logistic Regression

From Regression to Classification

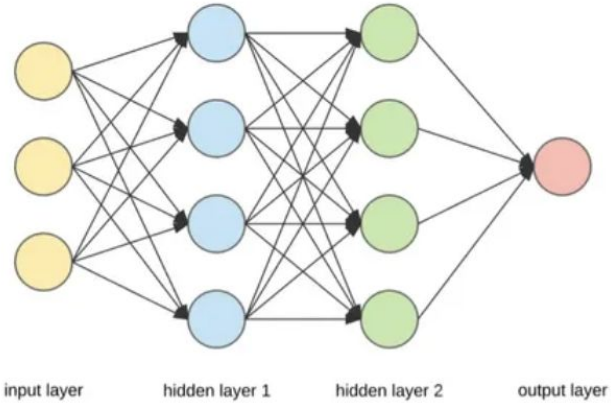


$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function; how?*



What’s its connection to Neural Networks

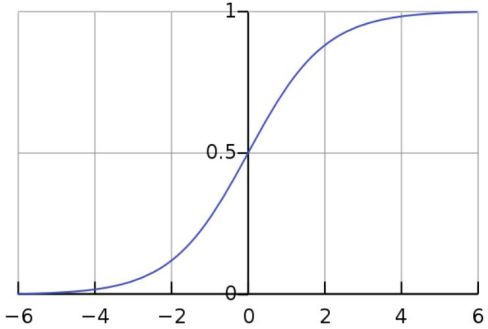


Logistic Regression

- Probabilistic classifier

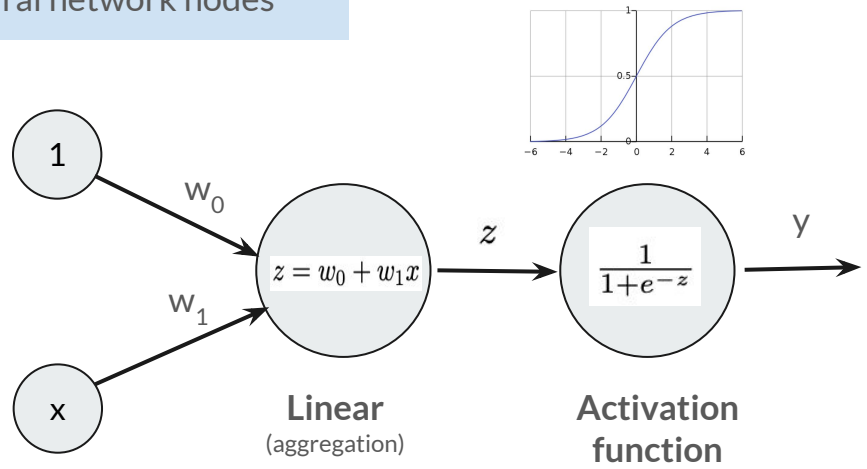
$$p(x) = \frac{1}{1+e^{-(w_0+w_1 x)}}$$

- Sigmoid function



Logistic Regression to Neural Networks (NNs)

Concept of neural network nodes

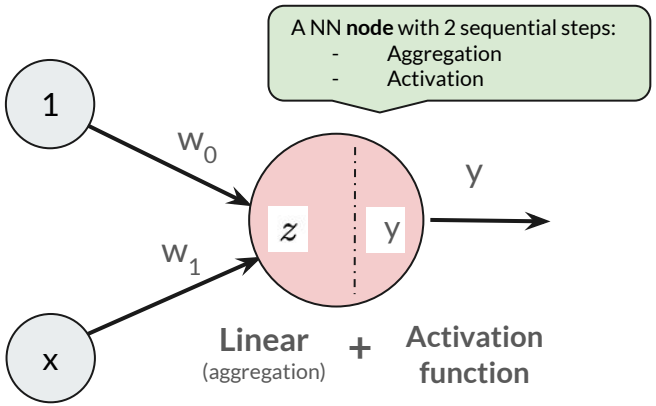


$y \in \{0, 1\}$

Logistic Regression

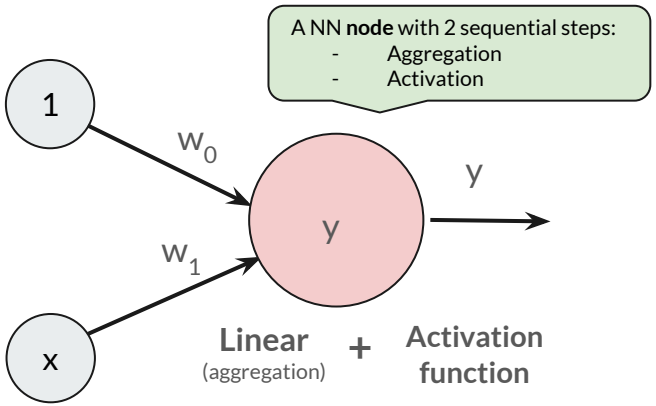
Neural Networks (Node)

Concept of neural network nodes



Neural Networks (Node)

Concept of neural network nodes



QA